WEB-BASED APPLICATION DEVELOPMENT

(CSE 3023)

<u>LAB REPORT 2</u>

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PROGRAMME	:	**COMPUTER SCIENCE (SOFTWARE ENGINEERING)**

Task 1

login.html

```html
<!DOCTYPE html>
<html lang="en">
    <head>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
        <title>Login Page</title>
    </head>
    <body>
        <h2>Login</h2>
        <form action="LoginServlet" method="POST">
            <label for="username">Username:</label>
            <input type="text" id="username" name="username"><br><br>
            <label for="password">Password:</label>
            <input type="password" id="password" name="password"><br><br>
            <input type="submit" value="Login">
        </form>
    </body>
</html>
```

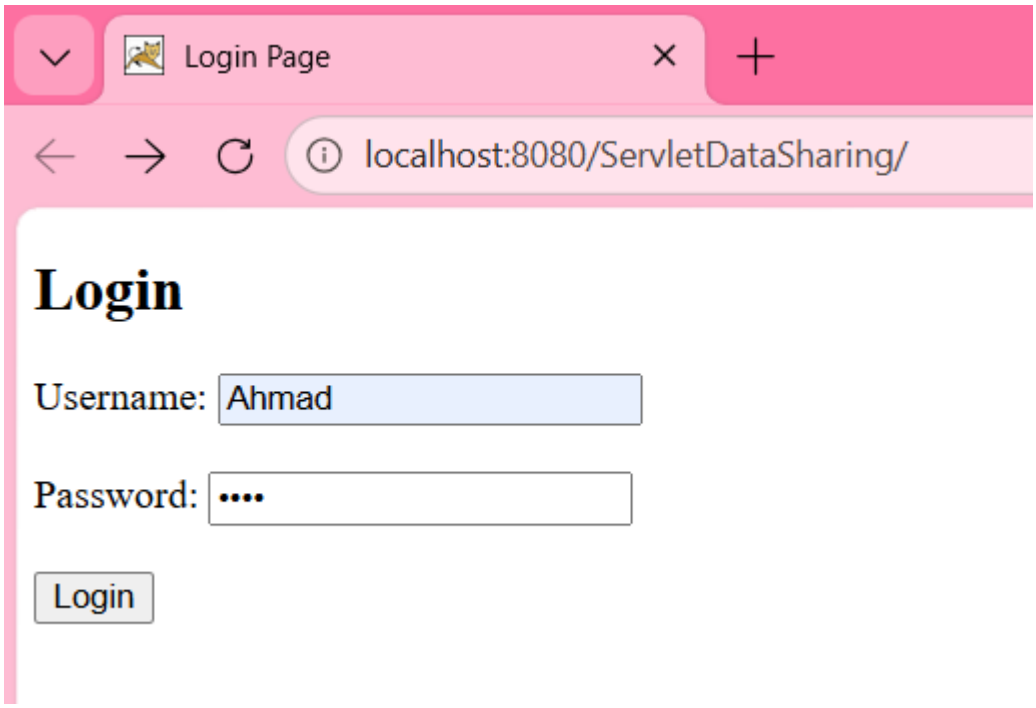
LoginServlet

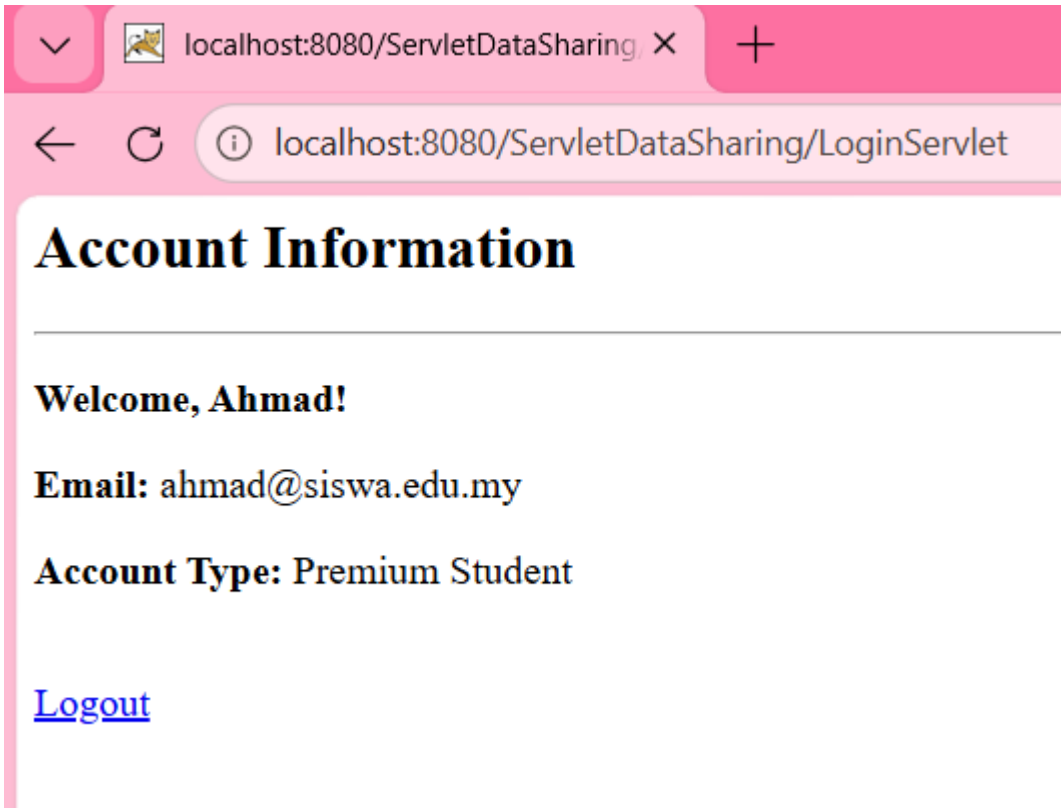
```java
protected void doPost(HttpServletRequest request, HttpServletResponse response) throws ServletException, IOException {
    String user = request.getParameter(name: "username");
    String pass = request.getParameter(name: "password");
    // Semakan kata laluan (Authentication)
    if (user.equals(anObject: "Ahmad") && pass.equals(anObject: "4567")) {
        // Data Sharing: Menyimpan maklumat tambahan ke dalam 'request'
        request.setAttribute(name: "accountType", o: "Premium Student");
        request.setAttribute(name: "email", o: "ahmad@siswa.edu.my");
        // Request Forwarding ke AccountServlet
        RequestDispatcher rd = request.getRequestDispatcher(path: "AccountServlet");
        rd.forward(request, response);
    } else if (user.equals(anObject: "Siti") && pass.equals(anObject: "1234")) {
        // Contoh untuk pengguna lain
        request.setAttribute(name: "accountType", o: "Standard Student");
        request.setAttribute(name: "email", o: "siti@siswa.edu.my");
        RequestDispatcher rd = request.getRequestDispatcher(path: "AccountServlet");
        rd.forward(request, response);
    } else {
        // Jika login gagal
        response.setContentType(type: "text/html");
        PrintWriter out = response.getWriter();
        out.println(x: "<h3 style='color:red'>Login Failed! Invalid credentials.< / h3 >");
        out.println(x: "<a href='login.html'>Try Again</a>");
    }
}
```

AccountServlet

```java
//Servlet AccountServle

import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

protected void doPost(HttpServletRequest request, HttpServletResponse response) throws ServletException, IOException {
    response.setContentType(type: "text/html");
    PrintWriter out = response.getWriter();
    // Mengambil semula parameter dari borang HTML asal
    String username = request.getParameter(name: "username");
    // Mengambil data yang dikongsi oleh LoginServlet
    String accountType = (String) request.getAttribute(name: "accountType");
    String email = (String) request.getAttribute(name: "email");
    out.println(x: "<html><body>");
    out.println(x: "<h2>Account Information</h2>");
    out.println(x: "<hr>");
    out.println("<p><strong>Welcome, " + username + "!</strong></p>");
    out.println("<p><b>Email:</b> " + email + "</p>");
    out.println("<p><b>Account Type:</b> " + accountType + "</p>");
    out.println(x: "<br>");
    out.println(x: "<a href='login.html'>Logout</a>");
    out.println(x: "</body></html>");
}
```

Output


Login Page
localhost:8080/ServletDataSharing/
Login
Username: Ahmad
Password: ••••
Login


localhost:8080/ServletDataSharing/
localhost:8080/ServletDataSharing/LoginServlet
Account Information
Welcome, Ahmad!
Email: ahmad@siswa.edu.my
Account Type: Premium Student
Logout

Task 2

```sql
USE CSE3023;

CREATE TABLE users (
    id INT AUTO_INCREMENT PRIMARY KEY,
    username VARCHAR(100),
    password VARCHAR(100),
    roles VARCHAR(50)
);
```

Output

←T→				id	username	password	roles
☐	Edit	Copy	Delete	2	abu	abu123	student
☐	Edit	Copy	Delete	3	ahmad	ahmad123	teacher
☐	Edit	Copy	Delete	4	ali	ali	student
☐	Edit	Copy	Delete	5	ali	ali123	student
☐	Edit	Copy	Delete	6	ali	ali123	student

Task 4

index.html

```html
<!DOCTYPE html>
<!--
Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
Click nbfs://nbhost/SystemFileSystem/Templates/JSP Servlet/Html.html to edit this template
-->
<html>
    <head>
        <title>TODO supply a title</title>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
    </head>
    <body>
        <h2>Login</h2>
        <form action="InsertServlet" method="POST">
            <label for="username">Username:</label>
            <input type="text" id="username" name="username"><br><br>
            <label for="password">Password:</label>
            <input type="password" id="password" name="password"><br><br>
            <label for="roles">Roles:</label>
            <input type="text" id="roles" name="roles"><br><br>
            <input type="submit" value="Login">
        </form>
    </body>
</html>
```

InsertServlet

```java
package com.lab.controller;

import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;

protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
    String user = request.getParameter(name: "username");
    String pass = request.getParameter(name: "password");
    String roles = request.getParameter(name: "roles");

    try {
        Class.forName(className:"com.mysql.cj.jdbc.Driver");
        Connection conn = DriverManager.getConnection(url:"jdbc:mysql://localhost:3306 / CSE3023", user: "root", password: "admin");

        String sql = "INSERT INTO users (username, password, roles)VALUES( ?,  ?,  ?)";
        PreparedStatement pstmt = conn.prepareStatement(string:sql);
        pstmt.setString(i: 1, string:user);
        pstmt.setString(i: 2, string:pass);
        pstmt.setString(i: 3, string:roles);
        pstmt.executeUpdate();
        pstmt.close();
        conn.close();
        response.sendRedirect(location: "ViewServlet");
    } catch (Exception e) {
        e.printStackTrace();
    }
}
```

ViewServlet

```java
package com.lab.controller;

import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.Statement;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;

import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
```

```java
protected void doGet(HttpServletRequest request, HttpServletResponse response) throws ServletException, IOException {

    response.setContentType(type: "text/html");
    PrintWriter out = response.getWriter();
    out.println(x: "<h2>User List</h2>");
    out.println(x: "<table border='1'><tr><th>ID</th><th>Username</th><th>Password </th><th>Role</th><th>Actions</th></tr>");

    try {
        Class.forName(className:"com.mysql.cj.jdbc.Driver");
        Connection conn = DriverManager.getConnection(url:"jdbc:mysql://localhost:3306/CSE3023", user: "root", password: "admin");

        Statement stmt = conn.createStatement();
        ResultSet rs = stmt.executeQuery(string:"SELECT * FROM users");

        while (rs.next()) {
            int id = rs.getInt(string:"id");
            out.println(x: "<tr>");
            out.println("<td>" + id + "</td>");
            out.println("<td>" + rs.getString(string:"username") + "</td>");
            // Untuk lab, kita paparkan password. Untuk sistem sebenar,jangan paparkan!
            out.println("<td>" + rs.getString(string:"password") + "</td>");
            out.println("<td>" + rs.getString(string:"roles") + "</td>");
            out.println("<td><a href='UpdateServlet?id=" + id + "'>Edit</a> | ");
            out.println("<a href='DeleteServlet?id=" + id + "'>Delete</a></td>");
            out.println(x: "</tr>");
        }
        out.println(x: "</table>");
        out.println(x: "<br><a href='index.html'>Add New User</a>");

        rs.close();
        stmt.close();
        conn.close();

    } catch (Exception e) {
        e.printStackTrace();
    }
}
```

DeleteServlet

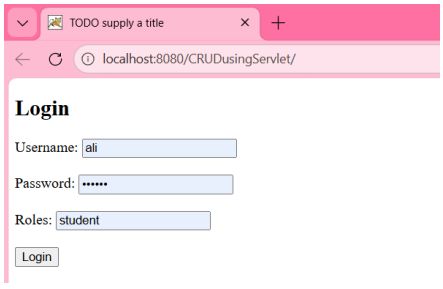
```java
protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

    String id = request.getParameter(name: "id");

    try {
        Class.forName(className:"com.mysql.cj.jdbc.Driver");
        Connection conn = DriverManager.getConnection(url:"jdbc:mysql://localhost:3306/CSE3023", user: "root", password: "admin");
        String sql = "DELETE FROM users WHERE id=?";

        PreparedStatement pstmt = conn.prepareStatement(string:sql);
        pstmt.setInt(i: 1, i1: Integer.parseInt(s: id));
        pstmt.executeUpdate();

        pstmt.close();
        conn.close();

        response.sendRedirect(location: "ViewServlet");
    } catch (Exception e) {
        e.printStackTrace();
    }
}
```

UpdateServlet

```java
protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
    response.setContentType(type: "text/html");
    PrintWriter out = response.getWriter();
    String id = request.getParameter(name: "id");

    try {
        Class.forName(className:"com.mysql.cj.jdbc.Driver");
        Connection conn = DriverManager.getConnection(url:"jdbc:mysql://localhost:3306/CSE3023", user: "root", password: "admin");

        String sql = "SELECT * FROM users WHERE id=?";
        PreparedStatement pstmt = conn.prepareStatement(string:sql);
        pstmt.setInt(i: 1, i1: Integer.parseInt(s: id));
        ResultSet rs = pstmt.executeQuery();

        if (rs.next()) {
            String currentRole = rs.getString(string:"roles");
            out.println(x: "<h2>Update User</h2>");
            out.println(x: "<form action='UpdateServlet' method='POST'>");
            out.println("<input type='hidden' name='id' value='" + rs.getInt(string:"id") + "'>");
            out.println("Username: <input type='text' name='username' value='" + rs.getString(string:"username") + "' required><br><br>");
            out.println("Password: <input type='text' name='password' value='" + rs.getString(string:"password") + "' required><br><br>");
            out.println(x: "Role: <select name='roles'>");
            out.println("<option value='Admin' " + (currentRole.equals(anObject: "Admin") ? "selected" : "") + ">Admin</option>");
            out.println("<option value='Staff' " + (currentRole.equals(anObject: "Staff") ? "selected" : "") + ">Staff</option>");
            out.println("<option value='Student' " + (currentRole.equals(anObject: "Student") ? "selected" : "") + ">Student</option>");
            out.println(x: "</select><br><br>");
            out.println(x: "<input type='submit' value='Update User'>");
            out.println(x: "</form>");
        }
        rs.close();
        pstmt.close();
        conn.close();
    } catch (Exception e) {
        e.printStackTrace();
    }
```

```java
protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

    String id = request.getParameter(name: "id");
    String username = request.getParameter(name: "username");
    String password = request.getParameter(name: "password");
    String roles = request.getParameter(name: "roles");

    try {
        Class.forName(className:"com.mysql.cj.jdbc.Driver");
        Connection conn = DriverManager.getConnection(url:"jdbc:mysql://localhost:3306/CSE3023", user: "root", password: "admin");

        String sql = "UPDATE users SET username=?, password=?, roles=?WHERE id=?";
        PreparedStatement pstmt = conn.prepareStatement(string:sql);
        pstmt.setString(i: 1, string:username);
        pstmt.setString(i: 2, string:password);
        pstmt.setString(i: 3, string:roles);
        pstmt.setInt(i: 4, i1: Integer.parseInt(s: id));
        pstmt.executeUpdate();

        pstmt.close();
        conn.close();

        response.sendRedirect(location: "ViewServlet");

    } catch (Exception e) {
        e.printStackTrace();
    }
}
```

Output



localhost:8080/CRUDusingServlet/

localhost:8080/CRUDusingServlet/ViewServlet

User List

ID	Username	Password	Role	Actions
3	ahmad	ahmad123	teacher	Edit | Delete
4	ali	ali	student	Edit | Delete
5	ali	ali123	student	Edit | Delete
6	ali	ali123	student	Edit | Delete
7	ali	ali123	student	Edit | Delete
8	ali	ali123	student	Edit | Delete
9	ali	ali123	Admin	Edit | Delete
10	ali	ali123	student	Edit | Delete
11	ali	ali123	student	Edit | Delete
12	ali	ali123	student	Edit | Delete
18	ali	ali123	student	Edit | Delete

Add New User

localhost:8080/CRUDusingServlet/

localhost:8080/CRUDusingServlet/UpdateServlet?id=3

Update User

Username: ahmad

Password: ahmad123

Role: Admin

Upda

Admin

Staff

Student

localhost:8080/CRUDusingServlet/

localhost:8080/CRUDusingServlet/ViewServlet

User List

ID	Username	Password	Role	Actions
4	ali	ali	student	Edit | Delete
6	ali	ali123	student	Edit | Delete
7	ali	ali123	student	Edit | Delete
8	ali	ali123	student	Edit | Delete
9	ali	ali123	Admin	Edit | Delete
10	ali	ali123	student	Edit | Delete
11	ali	ali123	student	Edit | Delete
12	ali	ali123	student	Edit | Delete

Add New User

Task 5

User.java

```java
package com.label.model;

/**
 *
 * @author chiaying
 */
public class User {

    private int id;
    private String username;
    private String password;
    private String roles;
// Constructor Kosong (Wajib untuk JavaBean)

    public User() {
    }
// Constructor dengan ID (Untuk Update & Delete)

    public User(int id, String username, String password, String roles) {
        this.id = id;
        this.username = username;
        this.password = password;
        this.roles = roles;
    }
// Constructor tanpa ID (Untuk Insert)

    public User(String username, String password, String roles) {
        this.username = username;
        this.password = password;
        this.roles = roles;
    }
// Getter dan Setter

    public int getId() {
        return id;
    }

    public void setId(int id) {
        this.id = id;
    }

    public String getUsername() {
        return username;
    }

    public void setUsername(String username) {
        this.username = username;
    }

    public String getPassword() {
        return password;
    }

    public void setPassword(String password) {
        this.password = password;
    }

    public String getRoles() {
        return roles;
    }

    public void setRoles(String roles) {
        this.roles = roles;
    }
}
```

UserDAO.java

```java
package com.lab.dao;

import com.label.model.User;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.util.ArrayList;
import java.util.List;

public class UserDAO {

    private String jdbcURL = "jdbc:mysql://localhost:3306/CSE3023";
    private String jdbcUsername = "root";
    private String jdbcPassword = "admin";
// Method untuk mendapatkan sambungan Database

    protected Connection getConnection() {
        Connection connection = null;
        try {
            Class.forName(className:"com.mysql.cj.jdbc.Driver");
            connection = DriverManager.getConnection(url:jdbcURL,
                    user: jdbcUsername, password: jdbcPassword);
        } catch (SQLException | ClassNotFoundException e) {
            e.printStackTrace();
        }
        return connection;
    }
// CREATE: Masukkan pengguna baharu

    public void insertUser(User user) {
        String sql = "INSERT INTO users (username, password, roles)VALUES( ?,  ?,  ?)";
        try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(string:sql)) {
            pstmt.setString(i: 1, string:user.getUsername());
            pstmt.setString(i: 2, string:user.getPassword());
            pstmt.setString(i: 3, string:user.getRoles());
```

```java
            pstmt.executeUpdate();
        } catch (SQLException e) {
            e.printStackTrace();
        }
    }
// READ: Ambil satu pengguna berdasarkan ID (Untuk Update)

    public User selectUser(int id) {
        User user = null;
        String sql = "SELECT id, username, password, roles FROM usersWHERE id =  ?";
        try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(string:sql)) {
            pstmt.setInt(i: 1, i1: id);
            ResultSet rs = pstmt.executeQuery();
            if (rs.next()) {
                String username = rs.getString(string:"username");
                String password = rs.getString(string:"password");
                String roles = rs.getString(string:"roles");
                user = new User(id, username, password, roles);
            }
        } catch (SQLException e) {
            e.printStackTrace();
        }
        return user;
    }
// READ: Ambil senarai semua pengguna (Untuk View)

    public List<User> selectAllUsers() {
        List<User> users = new ArrayList<>();
        String sql = "SELECT * FROM users";
        try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(string:sql)) {
            ResultSet rs = pstmt.executeQuery();
            while (rs.next()) {
                int id = rs.getInt(string:"id");
                String username = rs.getString(string:"username");
                String password = rs.getString(string:"password");
                String roles = rs.getString(string:"roles");
                users.add(new User(id, username, password, roles));
```

```java
            }
        } catch (SQLException e) {
            e.printStackTrace();
        }
        return users;
    }
// UPDATE: Kemaskini maklumat pengguna

    public boolean updateUser(User user) {
        boolean rowUpdated = false;
        String sql = "UPDATE users SET username = ?, password = ?,roles =  ? WHERE  id =  ?";
        try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(string:sql)) {
            pstmt.setString(i: 1, string:user.getUsername());
            pstmt.setString(i: 2, string:user.getPassword());
            pstmt.setString(i: 3, string:user.getRoles());
            pstmt.setInt(i: 4, i1: user.getId());
            rowUpdated = pstmt.executeUpdate() > 0;
        } catch (SQLException e) {
            e.printStackTrace();
        }
        return rowUpdated;
    }
// DELETE: Padam pengguna

    public boolean deleteUser(int id) {
        boolean rowDeleted = false;
        String sql = "DELETE FROM users WHERE id = ?";
        try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(string:sql)) {
            pstmt.setInt(i: 1, i1: id);
            rowDeleted = pstmt.executeUpdate() > 0;
        } catch (SQLException e) {
            e.printStackTrace();
        }
        return rowDeleted;
    }
}
```

InserServlet

```java
package com.lab.controller;

import com.lab.dao.UserDAO;
import com.label.model.User;
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;

public class InsertServlet extends HttpServlet {

    private UserDAO userDAO;

    public void init() {
        userDAO = new UserDAO();
    }

protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
    String user = request.getParameter(name: "username");
    String pass = request.getParameter(name: "password");
    String roles = request.getParameter(name: "roles");

    User newUser = new User(username: user, password: pass, roles);
    userDAO.insertUser(user: newUser);//Panggil DAO, bukan JDBC

    response.sendRedirect(location: "ViewServlet");
}
```

ViewServlet

```java
package com.lab.controller;

import com.lab.dao.UserDAO;
import com.label.model.User;
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.Statement;
import java.util.List;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;

import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

public class ViewServlet extends HttpServlet {

    private UserDAO userDAO;

    public void init() {
        userDAO = new UserDAO();
    }

protected void doGet(HttpServletRequest request, HttpServletResponse response) throws ServletException, IOException {

    response.setContentType(type: "text/html");
    PrintWriter out = response.getWriter();

    List<User> listUser = userDAO.selectAllUsers();//Panggil DAO

    out.println(x: "<h2>User List (Using DAO)</h2>");
    out.println(x: "<table border='1'><tr><th>ID</th><th>Username</th><th>Password </th><th>Role</th><th>Actions</th></tr>");

    for (User user : listUser) {
        out.print(s: "<tr>");
        out.print("<td>" + user.getId() + "</td>");
        out.print("<td>" + user.getUsername() + "</td>");
        out.print("<td>" + user.getPassword() + "</td>");
        out.print("<td>" + user.getRoles() + "</td>");
        out.print("<td><a href='UpdateService?id=" + user.getId() + "'>Edit</a> | ");
        out.print("<a href='DeleteServlet?id=" + user.getId() + "'>Delete</a></td>");
        out.println(x: "</tr>");
    }

    out.println(x: "</table>");
    out.println(x: "<br><a href='index.html'>Add New User</a>");

}
```

DeleteServlet

```java
package com.lab.controller;

import com.lab.dao.UserDAO;
import com.label.model.User;
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;

import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

public class DeleteServlet extends HttpServlet {

    private UserDAO userDAO;

    public void init() {
        userDAO = new UserDAO();
    }
```

```java
protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

    int id = Integer.parseInt(s: request.getParameter(name: "id"));
    userDAO.deleteUser(id); // Panggil DAO
    response.sendRedirect(location: "ViewServlet");

}
```

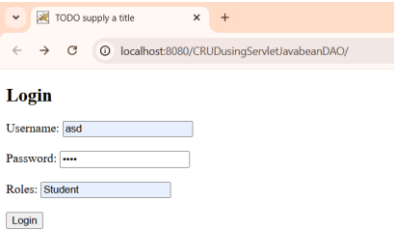
UpdateServlet

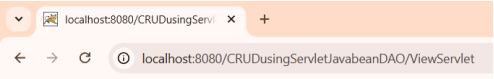
```java
package com.lab.controller;

import com.lab.dao.UserDAO;
import com.label.model.User;
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import java.sql.ResultSet;

import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

public class UpdateServlet extends HttpServlet {

    private UserDAO userDAO;

    public void init() {
        userDAO = new UserDAO();
    }
```

```java
protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
    response.setContentType(type: "text/html");
    PrintWriter out = response.getWriter();
    int id = Integer.parseInt(s: request.getParameter(name: "id"));

    User existingUser = userDAO.selectUser(id); // Panggil DAO

    if (existingUser != null) {
        out.println(x: "<h2>Update User (Using DAO)</h2>");
        out.println(x: "<form action='UpdateServlet' method='POST'>");
        out.println("<input type='hidden' name='id' value='" + existingUser.getId() + "'>");
        out.println("Username: <input type='text' name='username' value='" + existingUser.getUsername() + "' required><br><br>");
        out.println("Password: <input type='text' name='password' value='" + existingUser.getPassword() + "' required><br><br>");

        String currentRole = existingUser.getRoles();
        out.println(x: "Role: <select name='roles'>");
        out.println("<option value='Admin' " + (currentRole.equals(anObject: "Admin") ? "selected" : "") + "> Admin</option>");
        out.println("<option value='Staff' " + (currentRole.equals(anObject: "Staff") ? "selected" : "") + "> Staff</option>");
        out.println("<option value='Student' " + (currentRole.equals(anObject: "Student") ? "selected" : "") + "> Student</option>");
        out.println(x: "</select><br><br>");

        out.println(x: "<input type='submit' value='Update User'>");
        out.println(x: "</form>");
    }
}
```

```java
protected void doPost(HttpServletRequest request, HttpServletResponse response) throws ServletException, IOException {
    int id = Integer.parseInt(s: request.getParameter(name: "id"));
    String username = request.getParameter(name: "username");
    String password = request.getParameter(name: "password");
    String roles = request.getParameter(name: "roles");

    User user = new User(id, username, password, roles);
    userDAO.updateUser(user); // Panggil DAO
    response.sendRedirect(location: "ViewServlet");
}
```

Output





User List (Using DAO)

ID	Username	Password	Role	Actions
4	ali	ali	student	Edit | Delete
6	ali	ali123	student	Edit | Delete
7	ali	ali123	student	Edit | Delete
8	ali	ali123	student	Edit | Delete
9	ali	ali123	Admin	Edit | Delete
10	ali	ali123	student	Edit | Delete
11	ali	ali123	student	Edit | Delete
12	ali	ali123	student	Edit | Delete
19	asd	1235	Student	Edit | Delete

Add New User

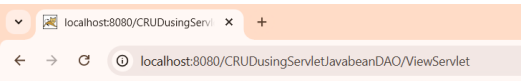
Update User (Using DAO)

Username: poll

Password: poll123

Role: Admin

Update User



User List (Using DAO)

ID	Username	Password	Role	Actions
4	ali	ali	student	Edit | Delete
6	poll	poll123	Admin	Edit | Delete
7	ali	ali123	student	Edit | Delete
8	ali	ali123	student	Edit | Delete
9	ali	ali123	Admin	Edit | Delete
10	ali	ali123	student	Edit | Delete
11	ali	ali123	student	Edit | Delete
12	ali	ali123	student	Edit | Delete
19	asd	1235	Student	Edit | Delete

Add New User

User List (Using DAO)

ID	Username	Password	Role	Actions
4	ali	ali	student	Edit | Delete
6	poll	poll123	Admin	Edit | Delete
7	ali	ali123	student	Edit | Delete
9	ali	ali123	Admin	Edit | Delete
10	ali	ali123	student	Edit | Delete
12	ali	ali123	student	Edit | Delete
19	asd	1235	Student	Edit | Delete

Add New User

Lab Exercise

database

```sql
CREATE TABLE products (
    id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(100),
    category VARCHAR(100),
    price DECIMAL(9,2),
    quantity INT
);
```

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0007 seconds.)

```sql
CREATE TABLE products ( id INT PRIMARY KEY AUTO_INCREMENT, name VARCHAR(100), category VARCHAR(100), price DECIMAL(9,2), quantity INT );
```

[Edit inline] [Edit] [Create PHP code]

Product.java

```java
public class Product {

    private int id;
    private String name;
    private String category;
    private double price;
    private int quantity;
// Constructor Kosong (Wajib untuk JavaBean)

    public Product() {
    }
// Constructor dengan ID (Untuk Update & Delete)

    public Product(int id, String name, String category, double price, int quantity) {
        this.id = id;
        this.name = name;
        this.category = category;
        this.price = price;
        this.quantity = quantity;
    }
// Constructor tanpa ID (Untuk Insert)

    public Product(String name, String category, double price, int quantity) {
        this.name = name;
        this.category = category;
        this.price = price;
        this.quantity = quantity;
    }
// Getter dan Setter

    public int getId() {
        return id;
    }

    public void setId(int id) {
        this.id = id;
    }

    public String getUsername() {
        return name;
    }

    public void setUsername(String name) {
        this.name = name;
    }

    public String getCategory() {
        return category;
    }

    public void setCategory(String category) {
        this.category = category;
    }

    public double getPrice() {
        return price;
    }

    public void setPrice(double price) {
        this.price = price;
    }

    public int getQuantity() {
        return quantity;
    }

    public void setQuantity(int quantity) {
        this.quantity = quantity;
    }
}
```

ProductDAO.java

```java
package com.lab.dao;

import com.label.model.Product;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.util.ArrayList;
import java.util.List;

/**
 *
 * @author chiaying
 */
public class ProductDAO {

    private String jdbcURL = "jdbc:mysql://localhost:3306/CSE3023";
    private String jdbcUsername = "root";
    private String jdbcPassword = "admin";
// Method untuk mendapatkan sambungan Database

    protected Connection getConnection() {
        Connection connection = null;
        try {
            Class.forName(className:"com.mysql.cj.jdbc.Driver");
            connection = DriverManager.getConnection(url:jdbcURL, user:jdbcUsername, password:jdbcPassword);
        } catch (SQLException | ClassNotFoundException e) {
            e.printStackTrace();
        }
        return connection;
    }
// CREATE: Masukkan pengguna baharu

    public void insertProduct(Product product) {
        String sql = "INSERT INTO products (name, category, price, quantity) VALUES(?, ?,  ?,  ?)";
        try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(string:sql)) {
```

```java
            pstmt.setString(i: 1, string:product.getName());
            pstmt.setString(i: 2, string:product.getCategory());
            pstmt.setDouble(i: 3, d: product.getPrice());
            pstmt.setInt(i: 4, i1: product.getQuantity());
            pstmt.executeUpdate();
        } catch (SQLException e) {
            e.printStackTrace();
        }
    }
// READ: Ambil satu pengguna berdasarkan ID (Untuk Update)

    public Product selectProduct(int id) {
        Product product = null;
        String sql = "SELECT id, name, category, price, quantity FROM products   WHERE id =  ?";
        try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(string:sql)) {
            pstmt.setInt(i: 1, i1: id);
            ResultSet rs = pstmt.executeQuery();
            if (rs.next()) {
                String name = rs.getString(string:"name");
                String category = rs.getString(string:"category");
                Double price = rs.getDouble(string:"price");
                int quantity = rs.getInt(string:"quantity");
                product = new Product(id, name, category, price, quantity);
            }
        } catch (SQLException e) {
            e.printStackTrace();
        }
        return product;
    }
// READ: Ambil senarai semua pengguna (Untuk View)

    public List<Product> selectAllProduct() {
        List<Product> products = new ArrayList<>();
        String sql = "SELECT * FROM products";
        try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(string:sql)) {
            ResultSet rs = pstmt.executeQuery();
            while (rs.next()) {
```

```java
                int id = rs.getInt(string:"id");
                String name = rs.getString(string:"name");
                String category = rs.getString(string:"category");
                Double price = rs.getDouble(string:"price");
                int quantity = rs.getInt(string:"quantity");
                products.add(new Product(id, name, category, price, quantity));
            }
        } catch (SQLException e) {
            e.printStackTrace();
        }
        return products;
    }
// UPDATE: Kemaskini maklumat pengguna

    public boolean updateProduct(Product product) {
        boolean rowUpdated = false;
        String sql = "UPDATE products SET name = ?, category = ?, price =  ?, quantity = ? WHERE  id =  ?";
        try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(string:sql)) {
            pstmt.setString(i: 1, string:product.getName());
            pstmt.setString(i: 2, string:product.getCategory());
            pstmt.setDouble(i: 3, d: product.getPrice());
            pstmt.setInt(i: 4, i1: product.getQuantity());
            pstmt.setInt(i: 5, i1: product.getId());
            rowUpdated = pstmt.executeUpdate() > 0;
        } catch (SQLException e) {
            e.printStackTrace();
        }
        return rowUpdated;
    }
// DELETE: Padam pengguna

    public boolean deleteProduct(int id) {
        boolean rowDeleted = false;
        String sql = "DELETE FROM products WHERE id = ?";
        try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(string:sql)) {
            pstmt.setInt(i: 1, i1: id);
            rowDeleted = pstmt.executeUpdate() > 0;
```

```java
        } catch (SQLException e) {
            e.printStackTrace();
        }
        return rowDeleted;
    }
}
```

InsertProductServlet

```java
package com.lab.controller;

import com.lab.dao.ProductDAO;
import com.label.model.Product;
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

/**
 *
 * @author chiaying
 */

public class InsertProductServlet extends HttpServlet {

    private ProductDAO productDAO;

    public void init() {
        productDAO = new ProductDAO();
    }

    /**
     * Processes requests for both HTTP <code>GET</code> and <code>POST</code>
     * methods.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType(type: "text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {
```

```java
        /* TODO output your page here. You may use following sample code. */
        out.println(x: "<!DOCTYPE html>");
        out.println(x: "<html>");
        out.println(x: "<head>");
        out.println(x: "<title>Servlet InsertProductServlet</title>");
        out.println(x: "</head>");
        out.println(x: "<body>");
        out.println("<h1>Servlet InsertProductServlet at " + request.getContextPath() + "</h1>");
        out.println(x: "</body>");
        out.println(x: "</html>");
    }
}

// <editor-fold defaultstate="collapsed" desc="HttpServlet methods. Click on the + sign on the left to edit the code.">
/**
 * Handles the HTTP <code>GET</code> method.
 *
 * @param request servlet request
 * @param response servlet response
 * @throws ServletException if a servlet-specific error occurs
 * @throws IOException if an I/O error occurs
 */
@Override
protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
    processRequest(request, response);
}

/**
 * Handles the HTTP <code>POST</code> method.
 *
 * @param request servlet request
 * @param response servlet response
 * @throws ServletException if a servlet-specific error occurs
 * @throws IOException if an I/O error occurs
 */
@Override
```

```java
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        String name = request.getParameter(name: "name");
        String category = request.getParameter(name: "category");
        double price = Double.parseDouble(s: request.getParameter(name: "price"));
        int quantity = Integer.parseInt(s: request.getParameter(name: "quantity"));

        Product newProduct = new Product(name, category, price, quantity);
        productDAO.insertProduct(product: newProduct);//Panggil DAO, bukan JDBC

        response.sendRedirect(location: "ViewProductServlet");
    }

    /**
     * Returns a short description of the servlet.
     *
     * @return a String containing servlet description
     */
    @Override
    public String getServletInfo() {
        return "Short description";
    }// </editor-fold>
}
```

ViewProductServlet

```java
package com.lab.controller;

import com.lab.dao.ProductDAO;
import com.label.model.Product;
import java.io.IOException;
import java.io.PrintWriter;
import java.util.List;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

/**
 *
 * @author chiaying
 */
public class ViewProductServlet extends HttpServlet {

    private ProductDAO productDAO;

    public void init() {
        productDAO = new ProductDAO();
    }

    /**
     * Processes requests for both HTTP <code>GET</code> and <code>POST</code>
     * methods.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType(type: "text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {
```

```java
        /* TODO output your page here. You may use following sample code. */
        out.println(x: "<!DOCTYPE html>");
        out.println(x: "<html>");
        out.println(x: "<head>");
        out.println(x: "<title>Servlet ViewProductServlet</title>");
        out.println(x: "</head>");
        out.println(x: "<body>");
        out.println("<h1>Servlet ViewProductServlet at " + request.getContextPath() + "</h1>");
        out.println(x: "</body>");
        out.println(x: "</html>");
    }
}

// <editor-fold defaultstate="collapsed" desc="HttpServlet methods. Click on the + sign on the left to edit the code.">
/**
 * Handles the HTTP <code>GET</code> method.
 *
 * @param request servlet request
 * @param response servlet response
 * @throws ServletException if a servlet-specific error occurs
 * @throws IOException if an I/O error occurs
 */
@Override
protected void doGet(HttpServletRequest request, HttpServletResponse response) throws ServletException, IOException {

    response.setContentType(type: "text/html");
    PrintWriter out = response.getWriter();

    List<Product> listProduct = productDAO.selectAllProduct();//Panggil DAO

    out.println(x: "<h2>Product List (Using DAO)</h2>");
    out.println(x: "<table border='1'><tr><th>ID</th><th>Name</th><th>Category</th><th>Price</th><th>Qunatity</th><th>Actions</th></tr>");

    for (Product product : listProduct) {
        out.print(s: "<tr>");
        out.print("<td>" + product.getId() + "</td>");
        out.print("<td>" + product.getName() + "</td>");
```

```java
            out.print("<td>" + product.getCategory() + "</td>");
            out.print("<td>" + product.getPrice() + "</td>");
            out.print("<td>" + product.getQuantity() + "</td>");
            out.print("<td><a href='UpdateProductServlet?id=" + product.getId() + "'>Edit</a> | ");
            out.print("<a href='DeleteProductServlet?id=" + product.getId() + "'>Delete</a></td>");
            out.println(x: "</tr>");
        }
        out.println(x: "</table>");
        out.println(x: "<br><a href='add_product.html'>Add New Product</a>");
    }

    /**
     * Handles the HTTP <code>POST</code> method.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }

    /**
     * Returns a short description of the servlet.
     *
     * @return a String containing servlet description
     */
    @Override
    public String getServletInfo() {
        return "Short description";
    }// </editor-fold>
}
```

UpdateProductServlet

```java
package com.lab.controller;

import com.lab.dao.ProductDAO;
import com.label.model.Product;
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

/**
 *
 * @author chiaying
 */
public class UpdateProductServlet extends HttpServlet {

    private ProductDAO productDAO;

    public void init() {
        productDAO = new ProductDAO();
    }

    /**
     * Processes requests for both HTTP <code>GET</code> and <code>POST</code>
     * methods.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType(type: "text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {
            /* TODO output your page here. You may use following sample code. */
```

```java
        out.println(x: "<!DOCTYPE html>");
        out.println(x: "<html>");
        out.println(x: "<head>");
        out.println(x: "<title>Servlet UpdateProductServlet</title>");
        out.println(x: "</head>");
        out.println(x: "<body>");
        out.println("<h1>Servlet UpdateProductServlet at " + request.getContextPath() + "</h1>");
        out.println(x: "</body>");
        out.println(x: "</html>");
    }
}

// <editor-fold defaultstate="collapsed" desc="HttpServlet methods. Click on the + sign on the left to edit the code.">
/**
 * Handles the HTTP <code>GET</code> method.
 *
 * @param request servlet request
 * @param response servlet response
 * @throws ServletException if a servlet-specific error occurs
 * @throws IOException if an I/O error occurs
 */
@Override
protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
    response.setContentType(type: "text/html");
    PrintWriter out = response.getWriter();
    int id = Integer.parseInt(s: request.getParameter(name: "id"));

    Product existingProduct = productDAO.selectProduct(id); // Panggil DAO

    if (existingProduct != null) {
        out.println(x: "<h2>Update Product (Using DAO)</h2>");
        out.println(x: "<form action='UpdateProductServlet' method='POST'>");
        out.println("<input type='hidden' name='id' value='" + existingProduct.getId() + "'>");
        out.println("Name: <input type='text' name='name' value='" + existingProduct.getName() + "' required><br><br>");
        out.println("Category: <input type='text' name='category' value='" + existingProduct.getCategory() + "' required><br><br>");
        out.println("Price: <input type='text' name='price' value='" + existingProduct.getPrice() + "' required><br><br>");
```

```java
        out.println("Quantity: <input type='text' name='quantity' value='" + existingProduct.getQuantity() + "' required><br><br>");

        out.println(x: "<input type='submit' value='Update Product'>");
        out.println(x: "</form>");
    }
}

// Simpan data yang telah dikemaskini
protected void doPost(HttpServletRequest request, HttpServletResponse response) throws ServletException, IOException {
    int id = Integer.parseInt(s: request.getParameter(name: "id"));
    String name = request.getParameter(name: "name");
    String category = request.getParameter(name: "category");
    double price = Double.parseDouble(s: request.getParameter(name: "price"));
    int quantity = Integer.parseInt(s: request.getParameter(name: "quantity"));

    Product product = new Product(id, name, category, price, quantity);
    productDAO.updateProduct(product); // Panggil DAO
    response.sendRedirect(location: "ViewProductServlet");
}

/**
 * Returns a short description of the servlet.
 *
 * @return a String containing servlet description
 */
@Override
public String getServletInfo() {
    return "Short description";
}// </editor-fold>
}
```

DeleteProductServlet

```java
package com.lab.controller;

import com.lab.dao.ProductDAO;
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

/**
 *
 * @author chiaying
 */
public class DeleteProductServlet extends HttpServlet {

    private ProductDAO productDAO;

    public void init() {
        productDAO = new ProductDAO();
    }

    /**
     * Processes requests for both HTTP <code>GET</code> and <code>POST</code>
     * methods.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType(type: "text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {
            /* TODO output your page here. You may use following sample code. */
            out.println(x: "<!DOCTYPE html>");
```

```java
        out.println(x: "<html>");
        out.println(x: "<head>");
        out.println(x: "<title>Servlet DeleteServlet</title>");
        out.println(x: "</head>");
        out.println(x: "<body>");
        out.println("<h1>Servlet DeleteServlet at " + request.getContextPath() + "</h1>");
        out.println(x: "</body>");
        out.println(x: "</html>");
    }
}

// <editor-fold defaultstate="collapsed" desc="HttpServlet methods. Click on the + sign on the left to edit the code.">
/**
 * Handles the HTTP <code>GET</code> method.
 *
 * @param request servlet request
 * @param response servlet response
 * @throws ServletException if a servlet-specific error occurs
 * @throws IOException if an I/O error occurs
 */
@Override
protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

    int id = Integer.parseInt(s: request.getParameter(name: "id"));
    productDAO.deleteProduct(id); // Panggil DAO
    response.sendRedirect(location: "ViewProductServlet");

}

/**
 * Handles the HTTP <code>POST</code> method.
 *
 * @param request servlet request
 * @param response servlet response
 * @throws ServletException if a servlet-specific error occurs
 * @throws IOException if an I/O error occurs
```

```java
     */
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }

    /**
     * Returns a short description of the servlet.
     *
     * @return a String containing servlet description
     */
    @Override
    public String getServletInfo() {
        return "Short description";
    }// </editor-fold>
}
```

add_product.html

```html
<!DOCTYPE html>
<!--
Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
Click nbfs://nbhost/SystemFileSystem/Templates/JSP Servlet/Html.html to edit this template
-->
<html>
    <head>
        <title>TODO supply a title</title>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
    </head>
    <body>
        <h2>Add Product</h2>
        <form action="InsertProductServlet" method="POST">
            <label for="name">Name:</label>
            <input type="text" id="name" name="name"><br><br>
            <label for="category">Category:</label>
            <input type="text" id="category" name="category"><br><br>
            <label for="price">Price:</label>
            <input type="text" id="price" name="price"><br><br>
            <label for="price">Quantity:</label>
            <input type="text" id="quantity" name="quantity"><br><br>
            <input type="submit" value="Add Product">
        </form>
    </body>
</html>
```

Output

localhost:8080/CRUDusingServl

localhost:8080/CRUDusingServletJavabeanDAO/ViewProductServlet

Product List (Using DAO)

ID	Name	Category	Price	Qunatity	Actions
1	iio	ppp	123.0	1241	Edit | Delete
4	jjj	ppp	99.3	550	Edit | Delete
5	lllo	pp	12.3	5	Edit | Delete

Add New Product

localhost:8080/CRUDusingServl

localhost:8080/CRUDusingServletJavabeanDAO/UpdateProductServlet?id=1

Update Product (Using DAO)

Name: zzz

Category: lll

Price: 123.0

Quantity: 1241

Update Product

localhost:8080/CRUDusingServl

localhost:8080/CRUDusingServletJavabeanDAO/ViewProductServlet

Product List (Using DAO)

ID	Name	Category	Price	Qunatity	Actions
1	zzz	lll	123.0	1241	Edit | Delete
5	lllo	pp	12.3	5	Edit | Delete

Add New Product

TODO supply a title

localhost:8080/CRUDusingServletJavabeanDAO/add_product.html

Add Product

Name: wepp

Category: pepp

Price: 56.3

Quantity: 66

Add Product

localhost:8080/CRUDusingServl

localhost:8080/CRUDusingServletJavabeanDAO/ViewProductServlet

Product List (Using DAO)

ID	Name	Category	Price	Qunatity	Actions
1	zzz	lll	123.0	1241	Edit | Delete
5	lllo	pp	12.3	5	Edit | Delete
6	wepp	pepp	56.3	66	Edit | Delete

Add New Product